THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 13: **Two Variables, One Line** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $(5, 2)$ is a solution of $kx + 6y = 27$, then k is: 1

(a) 3 **(b)** 5 **(c)** 2 **(d)** 4

2. The pair of equations $x + 2y = 4$ and $4x + 6y = 1$ has: 1

(a) exactly one solution **(b)** no solution

(c) exactly two solutions **(d)** infinitely many solutions

3. A line with slope 3 passes through $(3, -3)$. Its y-intercept is: 1

(a) -13 **(b)** -12 **(c)** -11 **(d)** -10

4. The slope of the line $x = 4$ is: 1

(a) 1 **(b)** 4 **(c)** undefined **(d)** 0

5. Is the point $(4, -5)$ a solution of $6x + 4y = 4$? 1

(a) Only for $x = 0$ **(b)** No **(c)** Yes **(d)** Cannot be said

6. The x-intercept of the line $3x + y = 26$ is: 1

(a) $\frac{32}{3}$ **(b)** $\frac{23}{3}$ **(c)** $\frac{26}{3}$ **(d)** $\frac{29}{3}$

7. The slope of the line through $(2, -2)$ and $(-1, 7)$ is: 1

(a) -1 **(b)** -4 **(c)** -3 **(d)** -2

8. The slope of the line $2x + 4y = 6$ is: 1

(a) $-\frac{1}{2}$ **(b)** $\frac{1}{2}$ **(c)** $-\frac{3}{2}$ **(d)** $\frac{3}{2}$

9. If $(1, 4)$ is a solution of $kx + 4y = 23$, then k is: 1

(a) 6 **(b)** 9 **(c)** 7 **(d)** 8

10. The pair of equations $5x + 3y = 6$ and $20x + 12y = 27$ has: 1

(a) exactly one solution **(b)** infinitely many solutions

(c) exactly two solutions **(d)** no solution

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write $7x + 2y = 34$ in the form $y = mx + c$ and read off the slope and the y-intercept. 2

12. Find the points where the line $2x + 4y = 32$ cuts the x-axis and the y-axis. 2

13. If $(5, 0)$ is a solution of $kx + 4y = 20$, find k. 2

14. Write $3x - 5y = 8$ in the form $y = mx + c$ and read off the slope and the y-intercept. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find k if the points $(0, 3)$, $(2, 5)$ and $(4, k)$ lie on one straight line. 3

16. Find the equation of the line with slope -3 that passes through $(6, -3)$, in the form $y = mx + c$. 3

17. Find the slope of the line through $(5, 4)$ and $(7, -5)$. 3

OR

Find k if the points $(-2, 4)$, $(1, 13)$ and $(4, k)$ lie on one straight line. 3

18. Find the equation of the line with slope -5 that passes through $(6, -4)$, in the form $y = mx + c$. 3

OR

Find the slope of the line through $(5, 9)$ and $(4, -2)$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Draw the graph of $-4x + y = 0$ using a table of values for $x = -1, 0, 1, 2$. State its slope and where it cuts the y-axis. 5

OR

The relation between Fahrenheit F and Celsius C is $F = \frac{9}{5}C + 32$. Find F when $C = 20$, find C when $F = 212$, and say what the slope $\frac{9}{5}$ means. 5

20. Draw the graph of $-2x + y = -4$ using a table of values for $x = -1, 0, 1, 2$. State its slope and where it cuts the y-axis. 5

OR

The relation between Fahrenheit F and Celsius C is $F = \frac{9}{5}C + 32$. Find F when $C = 55$, find C when $F = 212$, and say what the slope $\frac{9}{5}$ means. 5

END OF QUESTION PAPER — Total: 40 Marks

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